

THE GREAT RESTRUCTURING

The Unilingual Intelligence: Are We Building an AI with a Monocultural Mind?

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A vast, uniform tree of knowledge: a metaphor for the limitations of a monocultural AI.

I grew up in a world shaped by the words of Solomon Deressa, the renowned Ethiopian poet, screenwriter, journalist, and essayist. His poetry was a landscape I inhabited, and I admired him not just as a masterful artist, but as a formidable public intellectual. Years later, I learned of his coaching practice and a technique he used that struck me with the force of revelation. When a client presented a complex problem, Solomon, a master of languages, would listen in English, then mentally translate it into Amharic for its piercing clarity, or into Afan Oromo for its rich, descriptive power. In the jump between these languages, he said, the problem's structure would shift, revealing solutions that were invisible moments before.



I've found this to be true in my own life. When I mentally translate a conversation or a story into Amharic, a different kind of understanding emerges. The emotional weight shifts, new connections appear, and the core of the matter reveals itself with a startling new perspective. This is not a quirk of a few multilingual individuals; it is a key to a deeper truth about the human mind. This raises an urgent question for our modern age: What happens when a machine that might one day guide human civilization is trained to reflect only a single worldview, as if one corner of the mind stood in for the whole?

This idea — that language molds thought — is known as linguistic relativity, or the Sapir-Whorf hypothesis. For decades, it was a subject of intense debate, but today, the evidence is compelling. Brilliant thinkers like cognitive scientist Lera Boroditsky have demonstrated its power through research, while philosophers like George Lakoff, in his classic book Metaphors We Live By, showed how language provides the very frameworks for our reasoning. Their work reveals that languages are different operating systems for the mind, each with its own unique genius.

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A Compass in the Mind: The Intelligence of Space

Take one of Lera Boroditsky's most famous examples: the Kuuk Thaayorre, an Aboriginal community in Australia. In English, we use egocentric coordinates rooted in our own bodies: "You have a bug on your *left* leg." In Kuuk Thaayorre, they use absolute cardinal directions: "You have a bug on your *south-southwest* leg." This isn't a choice; their language demands it. From a young age, their brains are trained to perform a constant, astonishing feat of orientation. This fosters a **worldview where the self is not the center of the universe**, but a single point within a larger, stable environmental grid, creating a profound and unbroken connection to place.

The AI Contrast: Today's language models can process geospatial data, but they have no internal compass — no sense of orientation. They exist as abstract networks, unanchored to place or point of view. They lack the intelligence of the Kuuk Thaayorre, an intelligence that understands that one's position relative to the landscape is a fundamental part of one's identity and knowledge.

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The Evidence in Our Words: The Intelligence of Truth

Linguistics also reveals how some languages build truth-checking directly into their grammar. In English, we can state, “He is chopping wood,” without revealing how we know this. We could have seen it, been told it, or are simply guessing. But in languages like Turkish or Quechua, the grammar itself forces you to specify the source of your knowledge. You must use a different verb ending if you saw the event with your own eyes versus if you heard about it from someone else, or if you inferred it from evidence. This system, called “evidentials,” builds a kind of epistemology directly into every sentence. It creates a cognitive habit of constantly tracking information sources, making its speakers more nuanced thinkers about the nature of evidence and truth.

The AI Contrast: Our current AI models struggle immensely with misinformation and source attribution — a problem we are desperately trying to fix with add-ons and patches. For an AI trained on a language with evidentials, this “truth-checking” would be a native, structural feature of its intelligence, not a bug to be fixed. It would be inherently more sophisticated about the reliability of information because its linguistic DNA would demand it.

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Who Gets the Blame? The Intelligence of Agency

Boroditsky also highlights another fascinating distinction in how languages handle agency. If you accidentally break a vase, an English speaker defaults to, “I broke the vase.” Our grammar pushes us toward finding a culprit. In Spanish or Japanese, it’s more natural to say, “The vase broke,” focusing on the event rather than the actor. This isn’t about evading responsibility; it reflects a worldview that is more attuned to context and circumstance. The societal implications are enormous. A culture that says “the vase broke” may be more inclined to look for systemic solutions — *why was the vase left in a precarious position?* — rather than simply assigning individual blame.

The AI Contrast: An AI trained predominantly on English might build a “just” system that is fundamentally punitive and individualistic. It could easily miss the wisdom of a more collectivist or systems-thinking approach, because its own grammar doesn’t naturally guide it there. It risks conflating an accident with an action, because the linguistic distinction is not as sharply defined.

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How Many Blues Do You See? The Intelligence of Perception

The same principle applies to our senses, as research on Russian speakers demonstrates. In English, we have one basic word: “blue.” In Russian, there are two distinct, common words for light blue (goluboy) and dark blue (siniy). They are not shades of the same color; to a Russian speaker, they are as different as red and pink are to us. This is not just trivia. Studies show that because their language has trained them to make this distinction from childhood, Russian speakers are measurably faster at differentiating between these shades. Their language has physically fine-tuned their sensory input.

The AI Contrast: An AI can identify billions of colors by their RGB or hex codes — a feat of mathematical precision. But this is not perception; it is data processing. It lacks the culturally and linguistically shaped *attention* that makes a certain shade of blue sacred, another royal, and another melancholy. It sees the pixels, but misses the meaning.

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The AI in the Room

These are not trivial differences; they are distinct forms of intelligence. And right now, AI is largely ignoring them. Training models on an English-dominated internet means we are building a mind saturated with one set of cultural assumptions. Even within English, dialects and culturally rooted expressions — from Black English Vernacular to Indian English — encode different ways of understanding the world, yet these too are often flattened by standardized training data.

We've seen this problem before. Early facial recognition systems, trained on insufficiently diverse datasets, failed disastrously when identifying non-white faces. We are now at risk of repeating this mistake on a cognitive level, creating an intelligence that is deaf to the polyphony of the human mind. It's a danger that a growing chorus of thinkers in fields like [AI ethics](#) and the "[decolonizing AI](#)" movement have urgently warned against. When deployed in global contexts, an AI that internalizes one culture's notion of blame, fairness, or time risks misalignment not just in language — but in ethics.

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A Strategic Future for AI

This insight is what drives my work today. I actively collaborate with various groups to accelerate the creation of large language models for local languages across the Global South. In these efforts, however, I consistently encounter a critical blind spot, especially among investors and leaders in the mainstream tech world. They tend to view these projects as charitable acts or niche conservation efforts, peripheral to the “real” work of advancing AI.

This perspective fundamentally misses the point. I don't think of these projects as simply supporting one country; I see them as unlocking a huge advantage for our collective global intelligence. Every language contains a unique blueprint for thinking. By supporting these initiatives, we are not just “saving” languages; we are forging new lenses to look at our most complex global problems. This is not philanthropy; it is a strategic investment in cognitive diversity.



A flourishing, multicolored tree of knowledge: representing the potential of a diverse, polyglot AI.

This demands a concrete, global effort. **What the Manhattan Project was for physics, a global polyglot AI initiative could be for cognition — an urgent and ambitious endeavor.** It requires funding open-source multilingual models, pioneering community-partnered data collection for low-resource languages, and developing new benchmarks that measure cultural and relational intelligence, not just logical reasoning.

The future of intelligence isn't about creating a single, perfect mind; it's about fostering a global chorus of them. It's about taking the profound lesson I learned from Solomon Deressa and embedding it into the very DNA of the machines that will shape our world.

Artificial Intelligence

Technology

Language

Ethiopia

Humanity